

Performance Testing

[illegible]

Maximum Marks: 5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual API/Browser testing; Postman or another load-testing tool can be used for formal performance measurements.
Type of Testing	Functional and basic response-time observation; formal load/stress testing pending.
Target Module / API	POST /generate; GET /health
Test Environment	Local development environment
Test Date	30 September 2026

Step 2: Test Scenarios

S.No	Scenario	Users	Expected Outcome	Status
1	GET /health	1	Returns status ok.	Pass/verify
2	Valid POST /generate	1	Returns generated document.	Pass/verify
3	Missing required input	1	Validation/error response.	Pass/verify
4	Backend unavailable from frontend	1	Friendly connection error.	Pass/verify
5	Export generated content	1	TXT/DOCX/PDF files download.	Pass/verify

Step 3: Performance Results

Metric	Target	Actual	Status
Average Response Time	< 2 sec	Not formally measured	To test
Maximum Response Time	< 5 sec	Not formally measured	To test
Throughput	Record during load test	Not measured	To test
Error Rate	< 1%	Not measured	To test
CPU Utilization	< 80%	Not measured	To test
Memory Utilization	< 80%	Not measured	To test

Step 4: Observations & Analysis

Generation performance depends on the Gemini API response and local environment. Formal load testing should be performed before production deployment; no unmeasured performance values are presented as actual results.